ADR IDENTIFICATION – BASICS

Pharmacovigilance Internship Task

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Course: Pharm D

Task: Identification of suspected Adverse Drug Reactions from patient case scenarios

Objective: Identify suspect drugs and correlate adverse-event timelines using clinical intake records.

Educational note: The cases in this report are structured educational case scenarios for demonstrating the Workora task workflow. They are not real patient records and should not be interpreted as clinical advice.

1. Task Objective

The purpose of ADR identification is to review a patient case, correlate medicine exposure with the timing of an adverse event, distinguish suspected medicines from co-medications, consider alternative causes, and document the suspected ADR in a structured format.

2. ADR Identification Approach

Each case was reviewed using the following sequence: **Patient** → **Drug exposure** → **Reaction** → **Temporal relationship** → **Alternative causes** → **Dechallenge/Rechallenge (if available)** → **Suspect drug classification** → **Documentation.**

Key criteria considered

Criterion	What was assessed
Temporal relationship	Whether the reaction occurred after exposure to the suspected medicine.
Known pharmacology	Whether the reaction is clinically/pharmacologically plausible for the medicine.
Alternative causes	Disease, infection, other medicines, or other explanations that could account for the event.
Dechallenge	Whether the reaction improved after withdrawal/reduction of the suspected medicine, when applicable.
Rechallenge	Whether the reaction reappeared after re-exposure, when applicable.
Co-medications	Other medicines taken during the relevant period that could also contribute.

Case 1: Case Analysis

Field	Case information
Patient	25-year-old female
Medical history	No significant comorbidity reported.
Suspected/primary medicine	Amoxicillin 500 mg orally three times daily for bacterial infection.
Co-medications	Paracetamol 500 mg as needed for fever.
Medication–reaction timeline	Day 0: Amoxicillin started. Day 2: Generalized erythematous skin rash and itching developed. Day 2: Amoxicillin dis
Clinical analysis	The reaction appeared after drug initiation with a plausible temporal relationship. The reaction improved after disco
Suspect drug	Amoxicillin
Identified ADR	Generalized skin rash with pruritus
Case classification	Suspected ADR; likely drug-related based on temporal relationship and improvement after withdrawal.

Case 2: Case Analysis

Field	Case information
Patient	58-year-old male
Medical history	Hypertension controlled on regular medication.
Suspected/primary medicine	Cisplatin-containing chemotherapy administered on Day 0.
Co-medications	Ondansetron and dexamethasone given as antiemetic/supportive therapy.
Medication–reaction timeline	Day 0: Chemotherapy administered. Evening of Day 0: Nausea began. Day 1: Recurrent vomiting. Day 2: Symptom
Clinical analysis	Nausea and vomiting occurred shortly after chemotherapy. The timing is compatible with chemotherapy-associated
Suspect drug	Cisplatin-containing chemotherapy regimen
Identified ADR	Nausea and vomiting
Case classification	Suspected chemotherapy-associated ADR; temporal association supports suspicion.

Case 3: Case Analysis

Field	Case information
Patient	42-year-old female
Medical history	Type 2 diabetes mellitus.
Suspected/primary medicine	Metformin 500 mg twice daily, newly started at Day 0.
Co-medications	Glimepiride 1 mg once daily, continued from before Day 0.
Medication–reaction timeline	Day 0: Metformin initiated. Day 3: Loose stools and abdominal discomfort began. Day 7: Symptoms persisted. Day
Clinical analysis	The gastrointestinal symptoms began after initiation of metformin and improved after it was withheld. Glimepiride ha
Suspect drug	Metformin
Identified ADR	Diarrhea with abdominal discomfort
Case classification	Suspected ADR; temporal relationship and positive dechallenge support the suspicion.

Case 4: Case Analysis

Field	Case information
Patient	67-year-old male
Medical history	Osteoarthritis and chronic gastritis.
Suspected/primary medicine	Ibuprofen 400 mg three times daily started for acute pain.
Co-medications	Pantoprazole 40 mg once daily.
Medication–reaction timeline	Day 0: Ibuprofen started. Day 5: Epigastric pain and heartburn increased. Day 6: Ibuprofen discontinued. Day 8: Sy
Clinical analysis	The gastrointestinal symptoms developed after initiation of ibuprofen and improved following discontinuation. Pre-e
Suspect drug	Ibuprofen
Identified ADR	Increased epigastric pain/heartburn
Case classification	Suspected NSAID-associated gastrointestinal ADR; alternative cause noted.

Case 5: Case Analysis

Field	Case information
Patient	34-year-old female
Medical history	No significant medical history.
Suspected/primary medicine	Azithromycin 500 mg once daily started for respiratory infection.
Co-medications	Cetirizine 10 mg at night.
Medication–reaction timeline	Day 0: Azithromycin started. Day 1: Palpitations and dizziness reported. Day 2: Azithromycin stopped after clinical r
Clinical analysis	The symptoms occurred soon after exposure and resolved after withdrawal. The underlying respiratory infection an
Suspect drug	Azithromycin
Identified ADR	Palpitations with dizziness
Case classification	Suspected ADR requiring further clinical/causality assessment.

3. Structured ADR Identification Summary

Case	Suspect drug	Reaction / ADR	Temporal relationship	Dechallenge	Alternative causes	Final identification
Case 1	Amoxicillin	Generalized skin rash with pruritus	Compatible	Positive	Other medicines; infection-related illness	Suspected ADR likely from scenario
Case 2	Cisplatin-containing chemotherapy regimen	Nausea and vomiting	Strongly compatible	Improved with supportive care	Underlying illness and concomitant medicines considered	Suspected ADR
Case 3	Metformin	Diarrhea with abdominal discomfort	Compatible	Positive	Glimepiride/underlying GI causes considered	Suspected ADR.
Case 4	Ibuprofen	Increased epigastric pain/heartburn	Compatible	Positive	Pre-existing gastritis is a possible contributor.	Suspected ADR.
Case 5	Azithromycin	Palpitations with dizziness	Compatible	Positive	Respiratory illness and co-medications considered.	Suspected ADR

4. Overall Findings

Across the educational cases, the suspected ADRs were identified by correlating the onset of symptoms with medication exposure, reviewing co-medications, considering alternative explanations, and examining the response after withdrawal when available. The cases demonstrate why ADR identification is a suspicion-generating process; a suspected ADR should undergo appropriate seriousness, severity, and causality assessment before regulatory conclusions are made.

5. Documentation Checklist

✓	Identifiable patient information recorded
✓	Suspected medicine documented with dose/route where available
✓	Reaction and onset documented
✓	Medication timeline correlated with reaction timeline
✓	Co-medications reviewed
✓	Alternative causes considered
✓	Dechallenge/rechallenge information documented when available
✓	Suspect drug and ADR summarized
✓	Case retained for further causality/seriousness assessment

6. Conclusion

ADR identification requires a structured review of patient history, medicine exposure, reaction characteristics, timing, co-medications, and alternative causes. In these educational cases, each suspected ADR was documented with a clear drug–event timeline and a reasoned identification of the suspected medicine. Further pharmacovigilance processing would include case validation, seriousness and severity assessment, causality assessment, coding, follow-up, and reporting as applicable.

7. References

1. World Health Organization. Safety monitoring of medicinal products: guidelines for setting up and running a Pharmacovigilance Centre. WHO.
2. Indian Pharmacopoeia Commission. Pharmacovigilance Programme of India (PvPI).
3. ICH E2A. Clinical Safety Data Management: Definitions and Standards for Expedited Reporting.
4. ICH E2D. Post-Approval Safety Data Management: Definitions and Standards for Individual Case Safety Reports.
5. Uppsala Monitoring Centre. Pharmacovigilance resources and individual case safety reporting.